IET ASSIGNMENT

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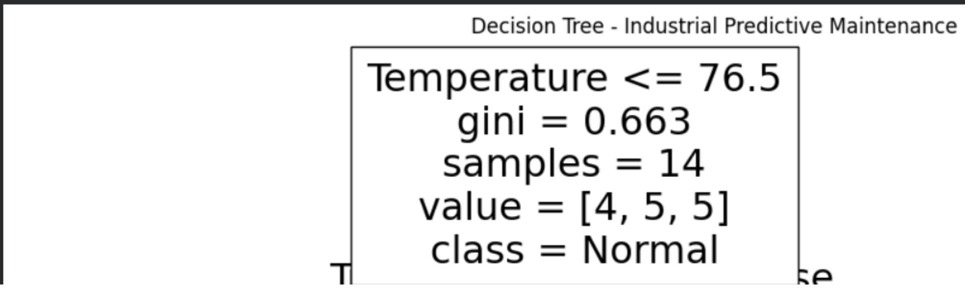
OUTPUTS

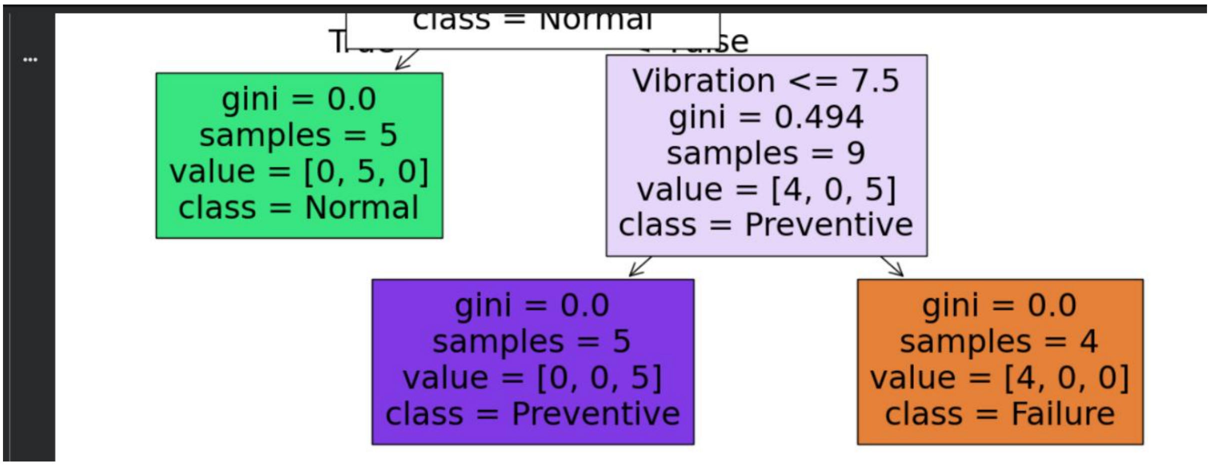
```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Industrial dataset
df = pd.DataFrame({
    "Temperature":[60,62,65,68,70,72,75,78,80,82,85,88,90,93,96,100,102,105,108,110],
    "Vibration":[1,2,2,3,3,4,4,5,5,5,6,6,7,7,8,8,9,9,10,11],
    "Speed":[1500,1520,1480,1510,1490,1500,1470,1450,1480,1460,
             1440,1420,1400,1380,1350,1320,1300,1280,1250,1200],
    "Torque":[35,38,40,42,45,47,50,52,54,55,58,60,62,65,68,70,72,75,78,80],
    "Hours":[500,800,1200,1600,2000,2500,3000,3500,4000,4500,
             5000,5500,6000,6500,7000,7500,8000,8500,9000,9500],
    "Condition":[
        "Normal","Normal","Normal","Normal","Normal",
        "Normal","Normal","Preventive","Preventive","Preventive",
        "Preventive","Preventive","Preventive","Preventive",
        "Failure","Failure","Failure","Failure","Failure","Failure"
    ]
})

# Input and target
X = df.drop("Condition", axis=1)
y = df["Condition"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=1, stratify=y
)

# Decision Tree
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    random_state=1
)

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Results
print("===== PREDICTIVE MAINTENANCE RESULTS =====")
print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

# Feature importance
print("\nFeature Importance:")
print(pd.DataFrame({
```

```
print(pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
}).sort_values("Importance", ascending=False))

# Decision Tree
plt.figure(figsize=(15,8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=model.classes_,
    filled=True
)
plt.title("Decision Tree - Industrial Predictive Maintenance")
plt.show()
```

```
===== PREDICTIVE MAINTENANCE RESULTS =====
Accuracy: 100.0 %

Confusion Matrix:
[[2 0 0]
 [0 2 0]
 [0 0 2]]

Classification Report:
              precision    recall  f1-score   support

     Failure       1.00      1.00      1.00         2
      Normal       1.00      1.00      1.00         2
  Preventive       1.00      1.00      1.00         2

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6


Feature Importance:
       Feature  Importance
0  Temperature    0.521368
1    Vibration    0.478632
2        Speed    0.000000
3       Torque    0.000000
4        Hours    0.000000
```




class = Normal
True
False
gini = 0.0
samples = 5
value = [0, 5, 0]
class = Normal
Vibration <= 7.5
gini = 0.494
samples = 9
value = [4, 0, 5]
class = Preventive
gini = 0.0
samples = 5
value = [0, 0, 5]
class = Preventive
gini = 0.0
samples = 4
value = [4, 0, 0]
class = Failure